

The Langlands Correspondence

Galois Side

Gal
 F

$\rho: \text{Gal} \rightarrow \text{GL}_n(\mathbb{Q}_\ell)$
continuous rep'n

$L(s, \rho) = \prod_v (1 - a_v p_v^{-s})^{-1}$
Euler product

bijection

Automorphic Side

GL_n
 (A_F)

π : automorphic rep
of $\text{GL}_n(A_F)$

$L(s, \pi) = \prod_v (1 - a_v p_v^{-s})^{-1}$
Euler product

Function Field Case (proved)

Drinfeld (1981, $n=2$) $\rightarrow$ Lafforgue (2002, all n) $\rightarrow$ Ngo (2009, Fundamental Lemma)